

Lab 1 – Data & Information

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1 Data vs Information

Databases organize data in order applying the business logic of the organization forming their data into information.

2 Data Models

Hierarchical databases store their data through a tree-like structure conveying information through relationships such as parent and child. Every child can only have a single parent and a parent can have an indefinite of children. These constraints nicely represent data that can form perfect trees. However it breaks down when any other relationship needs to be represented. For instance, let's go back to the Marist database imagine a simplistic version of storing sections, meetings, and their teachers. In a hierarchical database sections and meetings are a perfect match. At least simplistically, every section has 0 to many meetings and every meeting has exactly one parent. However, representing their teachers is difficult even if we simplify that each section only has a single teacher. Are they children of section? Then we would lose the information that children of sections are meetings. Are they children of meetings? This could work but then we would be storing professor data for each meeting. Whenever data is repeatedly stored it not only takes up more space but even more worryingly can lead to update, insertion, and deletion anomalies.

Maybe another type of data models can help us! Network databases permit a child to have more than one parent. This means we can have one instance of professor and all of their meetings point to them. We maintain the structure/

information that children of sections are meetings as well as children of meetings are professors without duplication. This is great until we have a professor that does not teach any meetings. Where should we put this professor? There is no good answer in either of these databases. This is one of the fundamental reasons why relational databases are largely thought as better.

There is another type of storage method known as XML. I personally have to appreciate XML as I use HTMX, a small javascript library which uses html attributes to send AJAX requests to an endpoint. I can represent all of my needs to send information from my frontend to my backend. However, I think it is better suited as a data transfer structure rather than a full data storage and query system because it runs into the same problems as network database. XML does allow multiple parents and children but still does not have a good place for data that is not connected.

3 Proof of Life

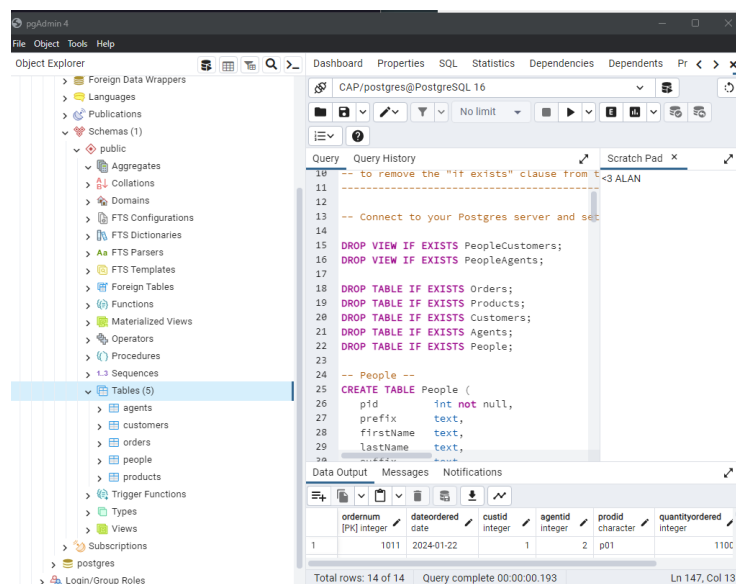


Figure 1: PG admin running on my computer showing the cap database

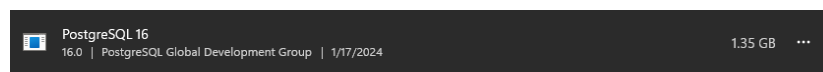


Figure 2: Postgre sql installed

```

→ psql --version
psql (PostgreSQL) 16.0
→ coding ⚡

```

Figure 3: psql cli tool working in powershell

```

1 {
2   "id": 164935,
3   "term": "202320",
4   "termDesc": "Spring 2023",
5   "courseReferenceNumber": "12114",
6   "partOfTerm": "1",
7   "courseNumber": "331L",
8   "subject": "CMPT",
9   "subjectDescription": "
10    Computing Technology",
11   "sequenceNumber": "111",
12   "campusDescription": "Marist
13    College Campus",
14   "scheduleTypeDescription": "
15    Lecture",
16   "courseTitle": "THEORY OF
17    PROGRAM LANG",
18   "maximumEnrollment": 25,
19   "enrollment": 22,
20   "seatsAvailable": 3,
21   "waitCapacity": 0,
22   "waitAvailable": 0,
23   "creditHourLow": 3,
24   "openSection": true,
25   "faculty": [
26     {
27       "bannerId": "38372",
28       "displayName": "Labouseur,
29        Alan",
30       "emailAddress": "Alan.
31        Labouseur@marist.edu",
32       "primaryIndicator": true
33     }
34   ],
35   "meetingsFaculty": [
36     {
37       "category": "01",
38       "meetingTime": {
39         "beginTime": "0800",
40         "building": "HC",
41         "buildingDescription": "
42          Hancock",
43         "campus": "AAA",
44         "campusDescription": "Marist
45          College Campus",
46         "category": "01",
47         "creditHourSession": 3.0,
48         "endDate": "05/12/2023",
49         "endTime": "0915",
50         "friday": false,
51         "hoursWeek": 1.25,
52         "meetingType": "LEC",
53         "meetingTypeDescription": "
54          Lecture",
55         "monday": false,
56         "room": "2005",
57         "saturday": false,
58         "startDate": "01/17/2023",
59         "sunday": false,
60         "thursday": false,
61         "tuesday": false,
62         "wednesday": true
63       }
64     },
65     {
66       "category": "01",
67       "meetingTime": {
68         "beginTime": "1100",
69         "building": "HC",
70         "buildingDescription": "
71          Hancock",
72         "campus": "AAA",
73         "campusDescription": "Marist
74          College Campus",
75         "category": "01",
76         "creditHourSession": 0.0,
77         "endDate": "05/12/2023",
78         "endTime": "1215",
79         "friday": true,
80         "hoursWeek": 1.25,
81         "meetingType": "LEC",
82         "meetingTypeDescription": "
83          Lecture",
84         "monday": false,
85         "room": "2005",
86         "saturday": false,
87         "startDate": "01/17/2023",
88         "sunday": false,
89         "thursday": false,
90         "tuesday": false,
91         "wednesday": false
92       }
93     }
94   ]
95 }

```

```
82 ],  
83 "sectionAttributes": []
```

```
84 }
```

Listing 1: A Course's section from
Banner's api